PIBC: Think like a housing economist

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Today's learning objectives

- Elementary financial analysis
  - Discounting
  - NPV and IRR
  - Cap rates
- A standard "monocentric" model of location
  - How price should change with location
  - Which land uses go where
- Supply elasticity, demand growth, and price growth
- Economists skepticism of affordable housing mandates
  - Doesn't mean they are bad
  - Do need a good reason for "in kind" vs cash

The time value of money

- \$1 now generally deemed more valuable than \$1 in the future
  - Inflation
  - Can invest the dollar now and earn a return
  - Impatience
- A discount rate translates future to present dollars
  - We will call this rate d
  - The "present value" of a payment of \$$x$ in t years is $\frac{x}{[1+d]^t}$.
  - Example if discount rate is 10% \$1,000 in 7 years has PV

$$\frac{1000}{[1.10]^7} \approx \$513$$

- Project Net Present Value ("NPV")
  - Is sum of PV associated with project at all dates
  - Including negative of investment today (date 0)

$$NPV = \sum_{t=0}^{\infty} x_t \, [1+d]^{-t}$$

  - (assuming constant discount rate "flat yield curve")

More on discounting

- Arguably, parents job to inculcate low discount rate
- IRR: "internal rate of return"
  - Like NPV Jeopardy
  - "At this discount rate, NPV of project is zero"
  - Positive IRR only if $IRR > d$
- Project pro formas
  - All the info needed to compute NPV and IRR
  - Much more detail than just dates and cash flows
  - Important ratio: "cap rate"
    - Net operating income (NOI) in year 1 divided by cost
      Acquisition cost for simple purchase
      Total project cost o/w
  - NOI vs cashflow
    - NOI ignores financing (interest and amortization)
    - Cashflow $\approx$ NOI after financing costs

The world's simplest pro forma

- Buy land for \$1M today
- Spend \$1M building something also today
- Sell for \$2.25M in one year
- Discount rate: 10%
- NPV: -1M - 1M + $\frac{2.5M}{1.1} \approx \$272K$
- IRR: -2M + 2.5M/(1+IRR) = 0 $\Rightarrow$ IRR=25%
- Note NPV and IRR fall if project takes 2 years vs 1: NPV: -2M + $\frac{2.5M}{1.1^2} \approx$ \$66K
- NPV higher if discount rate is lower, say 5% NPV: -2M + $\frac{2.5M}{1.05} \approx$ \$380K

Useful game to understand discounting

- Pick your favourite musician's last name
  - A-M: $d = 3\%$
  - N-Z: $d = 10\%$
- Match up with a partner
- Try to create a loan that has positive NPV to both of you
- Loan has amount L, interest rate r, and term of 1 year
- Borrower NPV: $L\left[1 - \frac{1+r}{1+d}\right]$
  - Positive iff $r < d$
- Lender NPV: $L\left[\frac{1+r}{1+d} - 1\right]$
  - Positive iff $r > d$
- If same discount rate, impossible to have $+$NPV for both

Segue: from game to why aforable housing mandates are dicey

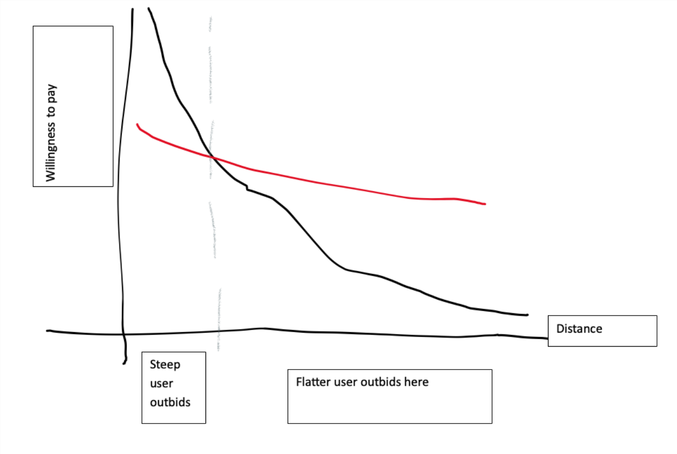
- What do markets do, ideally?
- Get everyone to agree on ratios of benefits to cost
- I have a discount rate of 10%, you 5%
    - You should lend me money at a rate between 5% and 10%
    - Positive NPV to both of us
- Very similar:
    - You can build an apartment for $500K
    - I value the apartment at $550K
- With "complete markets" trades happen until agreement
    - I have so much housing, I no longer value more than your cost
- Affordability mandates:
    - Someone willing to pay $400K
    - Market price $600K
    - Force sale at, say, $350K
    - Buyer gets +$50K, seller gets -$250K
    - "Pareto gain": seller gives buyer $150K!

The money value of time and urban land markets

- "Location, location, location"
- The classical "monocentric city model"
    - Everyone commutes to location 0: the CBD
    - Cost of commuting: t per unit of distance
    - How does land price change with distance?
- "Bid-rent curves"
    - Willingness to pay for land at distance x is $R(x)$
    - To be indifferent between living at x a tiny bit further
    - Consume $l(x)$ units of land
    - $R(x)l(x) = R(x+dx)l(x+dx) + t \cdot dx$
    - So slope of R curve is $-\frac{t}{l(x)}$
    - Land valuation falls with distance,
        - But more slowly if you consume more land
        - Hence steeper curve closer to downtown
        - (also t bigger downtown)

Equilibrium with multiple uses

- Remember, slope of bid-rent curve is $-\frac{t}{l(x)}$
- Closer to downtown will be steeper curve



Who should be closer to town given bid rent curve logic?

- Remember, slope of bid-rent curve is $-\frac{t}{l(x)}$
- Starbucks vs WalMart
- Yuppie versus family with 1 earner, 2 kids, 2 pets?
- Apartment building versus house with yard?
  - But zoning!
- Office building versus warehouse?

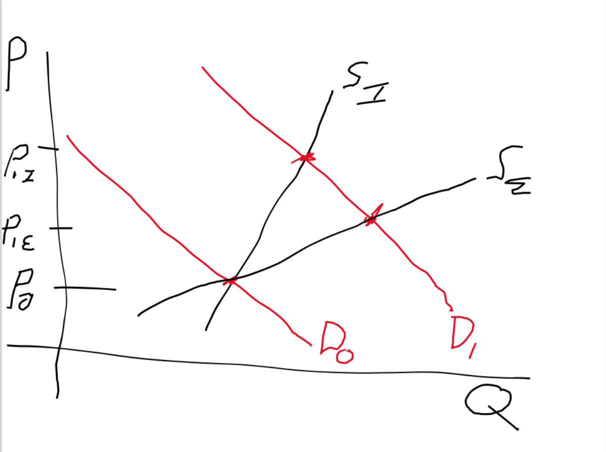
Where should cap rates be higher?

- Investors presumably get similar total returns
    - In all locations
    - In all asset classes
    - Otherwise too many net sellers or buyers
- If I buy, hold one period, then sell, "unleveraged" profit =

$$-\text{Price}_0 + \text{Price}_1 * \frac{caprate}{1+d} + \frac{\text{Price}_2}{[1+d]^2}$$

- So if return is equal, $\uparrow$ cap rate $\Rightarrow \downarrow \text{Price}_2$
- So higher cap rate in:
    - Vancouver (San Francisco) apartment 2021?
    - Toronto apartment 2025?
    - Winnipeg (Des Moines) shopping center?

Cap rates, elasticity of supply, and equilibrium
Where should cap rate be lower?
Inelastically supplied city if demand is growing



Recap of today's learning objectives

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